

EDUCATION

PhD in applied mathematics, *Université Paris-Saclay, INRIA, CentraleSupélec* Oct. 2021 - Jan. 2025

Divergence minimization with applications in variational inference, black-box global optimization, and adaptive importance sampling. Funded by ERC MAJORIS. Advised by E. Chouzenoux and V. Elvira.

MSc in applied mathematics, *Université Paris-Saclay* 2018 - 2020

M2 Optimization: optimal control, continuous optimization (theoretical and numerical aspects), stochastic optimization, game theory, calculus of variations, and tropical algebra.

Engineering degree, *ENSTA Paris, Institut Polytechnique de Paris* 2017 - 2020

Major in applied mathematics: discrete and continuous optimization, control theory, statistics, probability, dynamical systems, and partial differential equations.

POSITIONS

Postdoctoral researcher, *Statify team (INRIA, UGA), Grenoble, France* Jul. 2025 - Jul. 2026

Theoretical study of the convergence of stochastic natural gradient algorithms. With F. Forbes.

Research stays, *School of Mathematics (UoE), Edinburgh, Scotland* Feb. 2023 - May 2023, Mar. 2024

Exploration of the connections between variational inference and adaptive importance sampling, leading to novel adaptive importance sampling algorithms. With V. Elvira and N. Branchini.

Research engineer, *OPIS team (INRIA), Palaiseau, France* Dec. 2020 - Sep. 2021

Stochastic algorithms for non-convex optimization. Part of the project ERC MAJORIS. Advised by E. Chouzenoux and V. Elvira.

Research engineer, *LBE (INRAE), Narbonne, France* Oct. 2020 - Nov. 2020

Development of a Matlab code to simulate metabolic transitions at a finer scale in microbial populations. Part of the projects HME 3BCAR and ANR JANUS.

Master thesis, *INRAE, Montpellier, France* Apr. 2020 - Sep. 2020

Determination of the optimal periodic control for a scalar problem, with applications to the chemostat model and water bioremediation processes. Study of the multiple species case. Advised by A. Rapaport.

Research intern, *UTFSM, Valparaíso, Chile* May 2019 - Aug. 2019

Derivation of continuity properties and analysis of the sensitivity with respect to the initial conditions of the set of sustainable thresholds for a discrete-time controlled system. Advised by C. Hermosilla.

PUBLICATIONS

Journal papers

H. D. Nguyen, J. Westerhout, T. Guilmeau, and J. Arbel. **"Consistency of variational approximations under bounded Kullback-Leibler divergence"**. *Submitted*, 2026.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"Regularized Rényi divergence minimization through Bregman proximal gradient algorithms"**. *Journal of Machine Learning Research*, vol. 26(157), 2025.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"A divergence-based condition to ensure quantile improvement in black-box global optimization"**. *IEEE Transactions on Evolutionary Computation*, vol. 29(4), pp. 1017-1028, 2025.

P. Gajardo, T. Guilmeau, and C. Hermosilla. **"Sensitivity analysis of the set of sustainable thresholds"**. *Set-Valued and Variational Analysis*, vol. 32(18), 2024.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"On variational inference and maximum likelihood estimation with the λ -exponential family"**. *Foundations of Data Science*, vol. 6(1), pp. 85-123, 2024.

T. Guilmeau and A. Rapaport. **"Multiplicity of neutrally stable periodic orbits with coexistence in the chemostat subject to periodic removal rate"**. *SIAM Journal on Applied Mathematics*, vol. 84(1), pp. 39-59, 2024.

T. Guilmeau and A. Rapaport. **"Singular arcs in optimal periodic control problems with scalar dynamics and integral input constraint"**. *Journal of Optimization Theory and Applications*, vol. 195, pp. 953-975, 2022.

Conference papers

T. Guilmeau, H. Hendrikx, and F. Forbes. **"Convergence of projected stochastic natural gradient variational inference for various step size and sample or batch size schedules"**. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

T. Guilmeau, N. Branchini, E. Chouzenoux, and V. Elvira. **"Adaptive importance sampling for heavy-tailed distributions via α -divergence minimization"**. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"Adaptive simulated annealing through alternating Rényi divergence minimization"**. *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"Proximal-based adaptive simulated annealing for global optimization"**. *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

F. Dupeuble, A. Rapaport, T. Guilmeau, J. Tchouanti, B. Enjalbert, C. Bideaux, J.-P. Steyer, A. Feddaoui-Papin, and J. Harmand. **"Deterministic models to decipher the lag phase duration during diauxie"**. *IFAC-PapersOnLine*, 2022.

T. Guilmeau, E. Chouzenoux, and V. Elvira. **"Simulated annealing: a review and a new scheme"**. *Proceedings of the IEEE Statistical Signal Processing Workshop (SSP)*, 2021.

TEACHING

Teaching assistant, Numerical Optimisation - Université Grenoble Alpes

Spring 2026

A 4th year course on optimization, focusing on algorithms and their implementation. Part of the MSc Applied Mathematics of Université Grenoble Alpes.

Teaching assistant, Statistical Programming with R - Université Grenoble Alpes

Spring 2026

A 1st year course on the R language, focusing on descriptive statistics, data visualization, and data management. Part of the BUT Data Science of Université Grenoble Alpes.

Teaching assistant, STA101 - ENSTA Paris

Spring 2024

A 3rd year introductory course on statistics, focusing on parameter estimation and hypothesis testing.

Teaching assistant, AO101 - ENSTA Paris

Spring 2024

A 3rd year introductory course on optimization, with a special focus on the quadratic case and including algorithmic aspects.

Teaching assistant, OPT201 - ENSTA Paris

Fall 2023

A 4th year course on differentiable optimization, including optimality conditions, sub-differentiability, and duality theory.

Teaching assistant, Optimization - CentraleSupélec

Spring 2020, Spring 2021

A 4th year course covering linear and convex optimization, integer programming, and introducing some iterative algorithms.

GRANTS

Postdoctoral grant, MSTIC - Persyval, Université Grenoble Alpes

Jul. 2025 - Jul. 2026

I received funding for a one-year postdoctoral position during the MSTIC-Persyval 2025 call for projects. The application was jointly written with F. Forbes.

OUTREACH

RJMI, INRIA Paris

2022, 2024

The RJMI is a two-days research-based event aimed at girls in high school. I helped a group of students carry a research program over an open problem in mathematics.

Science fair, Université Paris-Saclay

2022, 2023

Scientific workshops open to the public are held over two days, and I helped animate the INRIA workshops about algorithms and cryptography.

Maths week, Académie de Créteil

2022

Presentation to high school students about my work, and how research is organized as a whole.

Documentary about AI, with students of BUT MMI

2022

These students have to create a documentary from scratch each year, including writing, filming, and editing. I introduced the students to artificial intelligence (this year's theme) and organized interviews with researchers.

OTHER ACADEMIC EXPERIENCES

Reviewing, Bernoulli - IEEE Transactions on Signal Processing - AISTATS 2026 - JMLR**MSc student supervision, INRIA Saclay**

2023

LANGUAGES

French: Native speaker**English:** Fluent (TOEIC: 990/990)**Spanish:** Intermediate level

CODING SKILLS

Scientific programming: I mostly code in Julia, but I regularly use Python and R as well.**Other:** $\text{\LaTeX}$, HTML, CSS.